

Data Flow Diagram

This document describes the data flow within OptiCrop at three levels: context level (Level 0), system level (Level 1), and module level (Level 2).

Level 0 Context Diagram

External Entities

Farmer (Primary User) Inputs 7 soil/climate parameters; receives crop recommendation

Researcher (Secondary) Views prediction logs; exports CSV for analysis

Admin Retrains model; updates crop metadata

Central Process: OptiCrop System Accepts user inputs Validates Scales features Runs ML prediction Returns crop recommendation with confidence score Logs to CSV

Level 1 System Data Flow Diagram

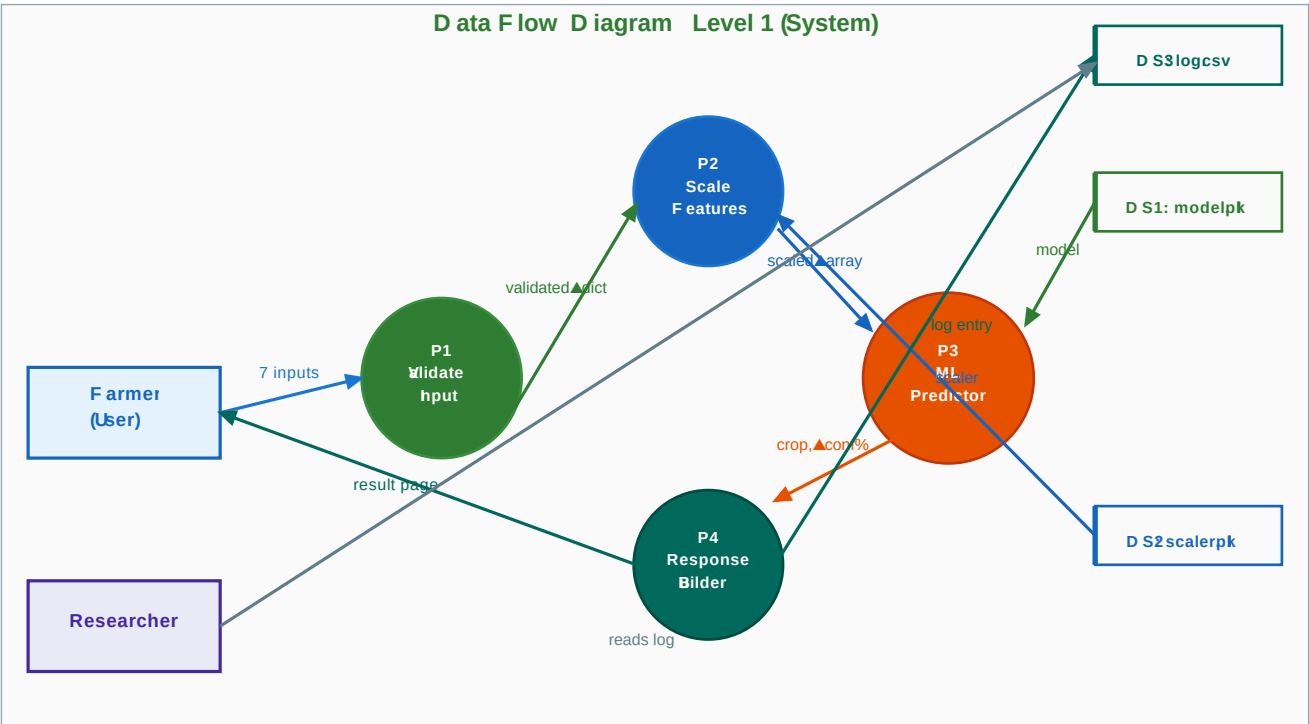


Figure 1: Level 1 Data Flow Diagram OptiCrop System

Level 1 Data Flow Reference Table

D	Data Flow	From	To	Content
1	User Input	Farmer	Input Validator	N, P, K, Temp, Humidity, pH, Rainfall (7 floats)

2	Validated Data	Input Validator	Feature Scaler	Clean dict of 7 numeric values
3	Scaled Array	Feature Scaler	ML Predictor	numpy array shape (1,7)
4	Prediction	ML Predictor	Response Builder	crop_label (str), confidence (float)
5	Result Page	Response Builder	Farmer	HTML result with crop name, %, top-3 list
6	Log Entry	Response Builder	Log Store (CSV)	Timestamp, inputs, output appended
7	Model File	Model Store (.pkl)	ML Predictor	Random Forest object loaded at startup
8	Scaler File	Scaler Store (.pkl)	Feature Scaler	StandardScaler loaded at startup

Data Stores

D	Name	Location	Description
1	model.pkl	models/	Trained Random Forest classifier read-only during inference
2	scaler.pkl	models/	Fitted StandardScaler read-only during inference
3	Crop_recommendation.csv	data/	Raw training dataset 2,200 rows x8 columns
4	predictions.csv	logs/	Append-only prediction log timestamp, inputs, output